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Magnetostratigraphy of the Moenkopi Formation at Bears Ears, Utah

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We have performed a paleomagnetic study on over 200 oriented samples collected from the Lower Triassic Moenkopi Formation at Bears Ears in southeastern Utah. Seven intervals of normal and reversed magnetic polarities are present at the Bears Ears site. We have correlated these magnetozones to a similar set established in the same formation 130 km to the northeast. At Bears Ears, magnetozone boundaries occur in different lithologic members of the Moenkopi Formation than corresponding magnetozone boundaries present in the section to the northeast. We conclude that contemporaneous lateral variations of sediment type and environement are a characteristic feature of the Moenkopi Formation.

INTRODUCTION

The Moenkopi Formation forms part of a thick sequence of red beds covering much of the Colorado Plateau in southeastern Utah and northeastern Arizona. *McKee* [1954], *Shoemaker and Newman* [1959], *Stewart et al.* [1972], and *Blakey* [1973] have presented detailed summaries of the lateral distribution of Moenkopi lithofacies over the entire plateau. The excellent natural exposures and relatively gentle dip of Moenkopi strata have made them ideal candidates for paleomagnetic study. *Helsley* [1969] and *Helsley and Steiner* [1974] have presented the results of such a study on a continuous section of the Moenkopi occurring in the salt anticline region in southwestern Colorado. They established the presence of at least 10 zones of normal and reversed polarity in that section.

The aim of the present study is to extend the paleomagnetic results of *Helsley and Steiner* [1974] to an additional section of the Moenkopi at Bears Ears, Utah. By comparing the magnetic directions in the two sections we hope to shed some light on the temporal relationship which exists between various members of the Moenkopi Formation as well as on the magnetic-recording process itself.

BEARS EARS GEOLOGY

The Bears Ears locality is situated on the southern edge of the Monument Upwarp in southeastern Utah, where the strata dip gently (less than 3°) northeast. The sampling locality is shown in Figure 1. Here the Moenkopi overlies the Organ Rock member of the Permian Cutler Formation and is in turn overlain by the Shinarump Conglomerate of the Chinle Formation.

The lowermost unit, the Hoskinnini member, consists of 14 m of a coarse massive reddish-brown sandstone, overlain by 8 m of an alternating siltstone-sandstone sequence. The base of the Hoskinnini appears to rest conformably on the underlying Organ Rock member of the Permian Cutler Formation, although the well-defined boundary between the two units suggests that a hiatus may be present. A distinctive 'crinkly' bed having small-wave stratification, which *Stewart* [1959] and *Blakey* [1973] have recognized within the Hoskinnini throughout most of southeastern Utah, occurs near to the top of the basal massive sandstone. No identifiable fossils have been found in the Hoskinnini, which makes both age and paleoenvironmental interpretations difficult. *O'Sullivan* [1965] has suggested a lacustrine environment based on the occurrence of contorted bedding and gypsum stringers throughout the unit.

The Hoskinnini is overlain by 7.5 m of weak reddish-brown shaley clay stone, containing a few platy sandstone layers at the base. The corresponding contact with the Hoskinnini is marked by a thin mud pebble conglomerate, which suggests that a local unconformity may be present. This unit has been termed the Lower Slope Forming member by *Stewart* [1959]. The Slope Forming member is overlaid by 47-m thick sequence of alternating sandstones, siltstones, and clay stone–mudstones. The sandstone units form prominent ledges and exhibit large-scale cross laminations, ripple laminae, and load casts. The interbedded clay stone and mudstone units are usually pale brown to reddish brown, thin-bedded, and some show small-scale cross laminations. *McKee* [1954] has reported plant impressions, bones of amphibians, and some possible fish remains within this sequence at the Bears Ears locality. Lithologically, the sequence corresponds to *Blakey*'s [1973] Hideout Canyon facies within the Ledge Forming member. Several of the massive sandstone and siltstone units contain mud pellet conglomerates near their bases, suggesting that local breaks in deposition have occurred. *Blakey* [1973] has concluded that the Hideout Canyon facies are typical of floodplain deposits representing the initial stage of long deltaic episode. He also suggests that part of the Ledge Forming member can be correlated with the Lower Triassic Thaynes and Ankareh Formations in the Wasatch Mountains.

The uppermost unit of the Moenkopi at Bears Ears consists of 13.6 m of a pale reddish-brown clay stone unit, termed the Upper Slope Forming member. In places the unit is light olive-gray in color and contains a bleached zone at the top. The unconformity with the overlying Shinarump Conglomerate of the Chinle Formation has a local relief of 0.6–0.9 m. No fossils have been reported in this unit, which makes age correlation with similar units in other areas difficult. However, the unit has a stratigraphic equivalent at nearly all locations where the Moenkopi outcrops. *Blakey* [1973] suggests that the Upper Slope Forming member was deposited on and adjacent

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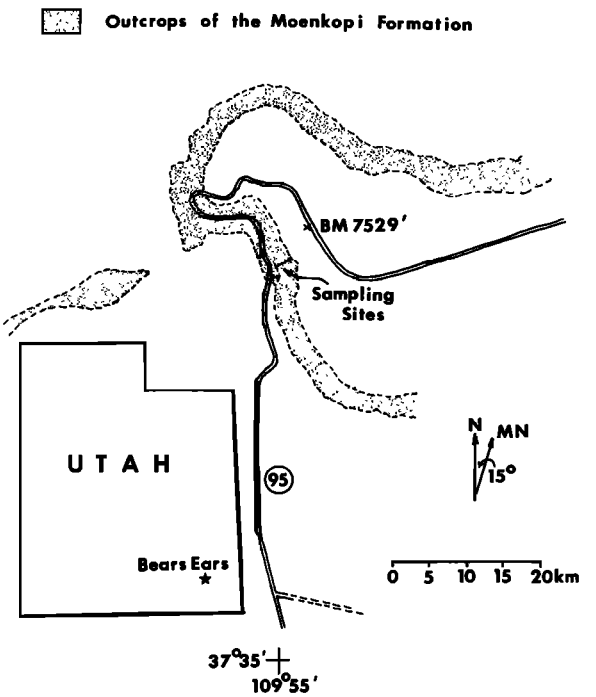


Fig. 1. Location map showing sampling localities and areas where the Moenkopi Formation outcrops.

to a protected shoreline, possibly in a sabkha environment, as described by *Kinsman* [1969].

Stewart [1959] has discussed the stratigraphic relationships between the individual members of the Moenkopi Formation defined in southeastern Utah and those in the salt anticline region to the northeast. He concludes that a major part of the lowermost member of the Moenkopi in the salt anticline region, the Tenderfoot member, is physically continuous with the Hoskinnini member. Additional units present in the Tenderfoot member were either not deposited in the southwest or may, as *Stewart* [1959] suggests, grade laterally into the Hoskinnini. The crinkly bed which is present in the Hoskinnini at Bears Ears has not been recognized in the salt anticline region.

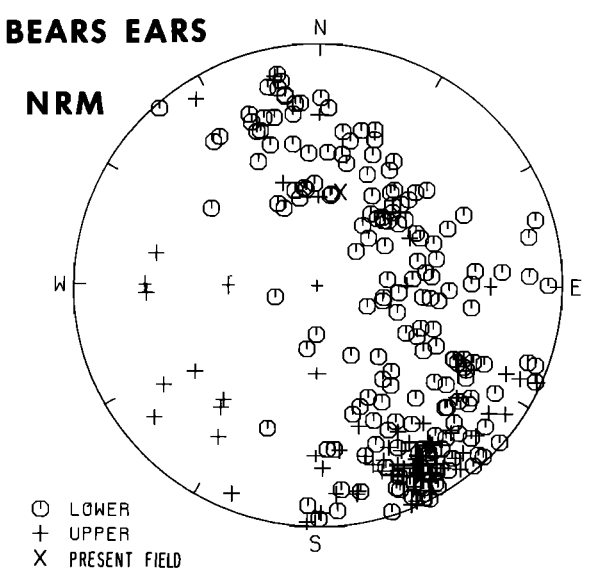


Fig. 2. Equal area projectional plot of NRM directions.

EXPERIMENTAL PROCEDURES

Over 200 oriented samples were collected from the section at an average stratigraphic interval of 36 cm. The 2.5-cm-diameter samples were drilled with a portable gasoline corer [*Helsley*, 1967] and were oriented using a Brunton Compass. The sampling localities are shown in Figure 1. The natural remanent magnetizations (NRM) of the samples were measured on either a PAR SM1 spinner magnetometer, or an SCT 6-cm access cryogenic magentometer. The noise limits of both these instruments are less than 5×10^{-4} A m^{-1} (5×10^{-9} emu/cm^3), and the angular accuracies are better than one degree. Thermal demagnetization experiments were performed using a noninductively wound furnace in which the residual magnetic field was kept at less than 10 nT, using mu-metal shields.

MAGNETIC RESULTS

An equal area projection plot of the NRM directions is shown in Figure 2. These form a prominent streak between both reversed and normal groupings and the present field. The antipodal groupings resemble the normal and reversed directions obtained by *Helsley and Steiner* [1974] near the Dolores River, Colorado. Shown in Figure 3 are the changes in intensity of magnetization which were observed in selected samples during stepwise thermal demagnetization. On the basis of the intensity changes the samples can be divided into two groups. The first group, which we shall term 'stable,' shows a slight decrease in intensity up to about 200°C, then remains fairly constant up to temperatures of 650°C, where the intensities fall off rapidly, indicating a Curie temperature close to that of hematite (675°C). The second group, termed 'unstable,' exhibits a more complex behavior, showing regular decreases in their intensities up to 650°C. In addition, three of the samples have a 'knee' close to 550°C, which may indicate the presence of magnetite (Curie point 578°C). Lithologically, the unstable samples were all coarser-grained sandstones, while the stable samples were generally shales or siltstones. A

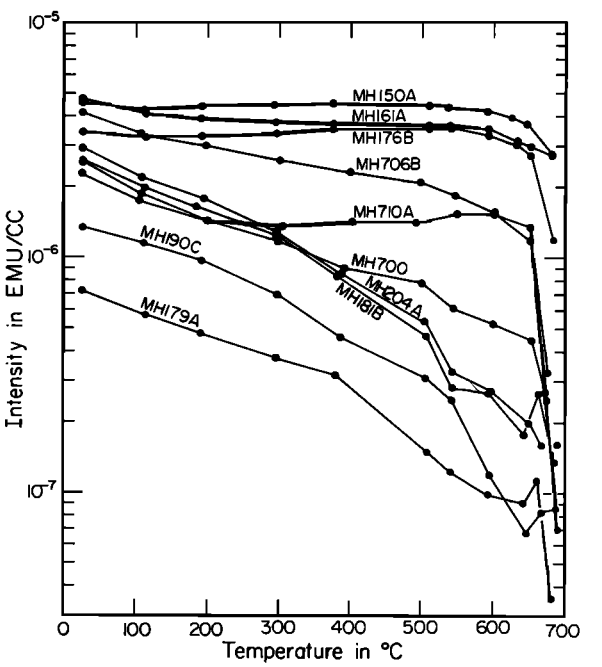


Fig. 3. Changes in intensity observed in the 10 pilot samples during stepwise thermal demagnetization.

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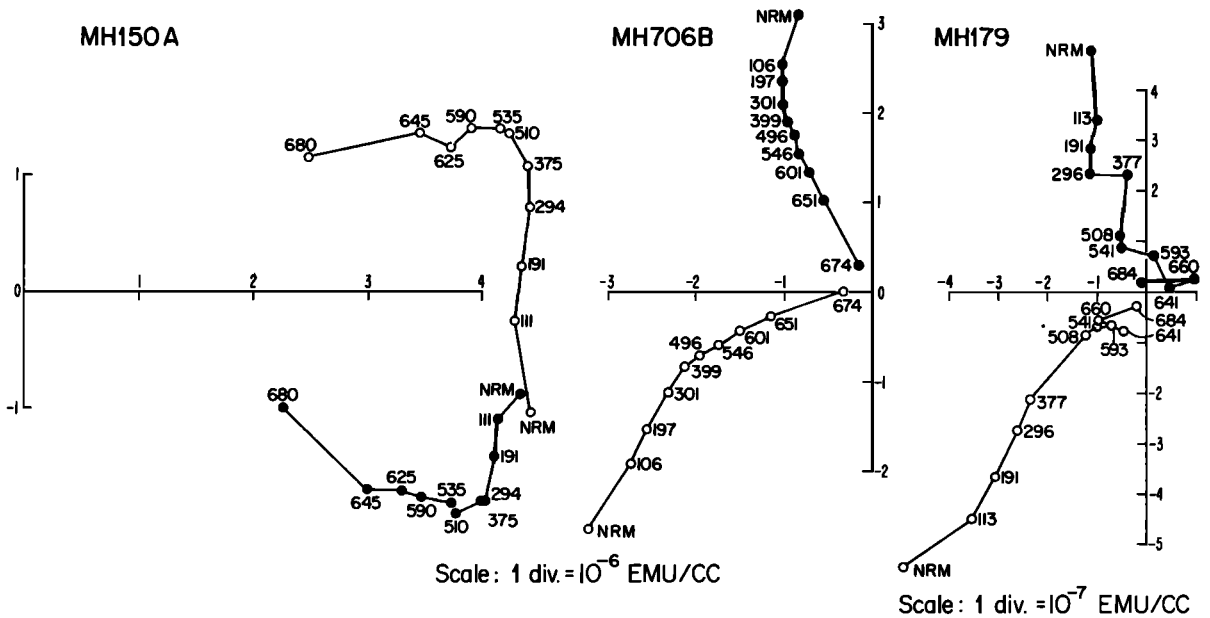


Fig. 4. Orthogonal demagnetization plots for three representative samples. The solid circles represent the directions and intensities of the two horizontal components of magnetization where the vertical axis is parallel to the present magnetic field direction. The open circles represent the total horizontal components (ordinates) versus the total vertical components (abscissae) of magnetization. The numbers beside each circle are the thermal demagnetization temperatures in degrees centigrade.

similar correspondence between paleomagnetic stability and grain size was noted by *Helsley* [1969] at Dolores, Colorado.

The directional changes observed in four of the samples are shown in Figure 4. Sample MH706B clearly reaches a stable normal direction at temperatures exceeding 550°C. The direction of the removed vector is fairly close to the present field direction at Bears Ears (decliniation is 15°, inclination is 55°). Samples MH150A and MH710A show a clear movement away from the present field direction toward a reversed direction which they reach at temperatures close to 550°C. The directions of the removed components are again close to that of the present field. The other stable samples (MH161A and MH176B) showed very similar behavior to samples MH710A and MH150A. Sample MH179 shows behavior which was characteristic of the unstable samples. Although there seems to be an initial trend away from the present field toward a reversed direction, no stable endpoint is reached.

On the basis of the demagnetization results we initially demagnetized all the samples at a temperature of 550°C. An equal area plot of these demagnetized directions appears in Figure 5. Most of the directions now cluster fairly well around two antipodal directions. However, a significant number of the directions are still neither reversed or normal. All samples showing this latter behavior were therefore demagnetized at 600°C and 650°C. These data were then examined to determine whether the directions of motion of the magnetic vectors were away from or toward a normal direction. Shown in Table 1 are the mean directions and statistical parameters calculated for the N2 and R3 intervals from the NRM and thermal demagnetization data at 550°C and 650°C. The N2 interval data show a clear movement away from the present field direction toward the mean Moenkopi normal direction, D = 345.7, I = 16.6, given by *Helsley* [1969, Table 7]. The R3 data,

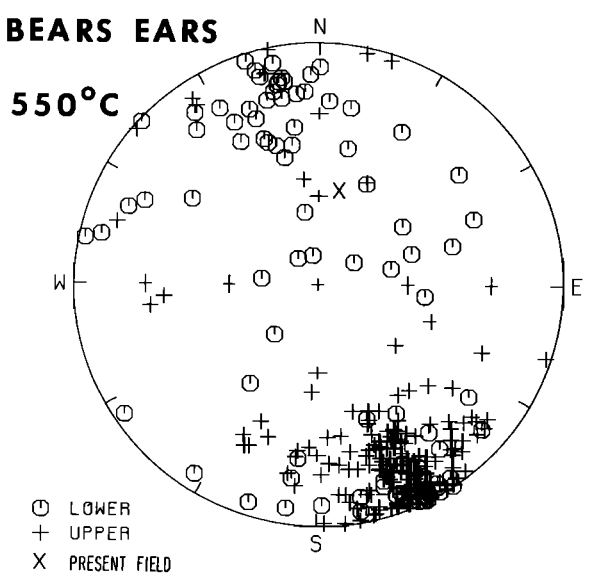


Fig. 5. Equal area projectional plot of sample magnetization directions after thermal demagnetization at temperatures between 500°C and 650°C.

TABLE 1. Changes in Mean Directions and Precision Parameters With Thermal Demagnetization

	D	I	α_{95}	k
	N1 Interval (N = 4)			
NRM	354.4	31.9	22.4	17.8
550°C	346.8	24.7	22.9	17.1
650°C	344.0	14.4	26.3	13.2
	R3 Interval (N = 17)			
NRM	175.1	66.0	42.1	1.7
550°C	195.4	−20.4	69.9	1.2
650°C	194.1	−29.1	61.1	1.3

D is the mean declination from geographic north, I is the mean inclination, α_{95} is the half angle of cone of 95% confidence, k is *Fisher's* [1953] precision parameter, and N is the number of samples.

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TABLE 2. Mean Directions and Statistical Parameters for Vectors Removed During Demagnetization at 550°C

Interval	N	D	I	α_{95}	k
R1	58	39.2	60.9	5.6	12.1
N1	4	7.2	42.4	28.4	11.4
R2	2	27.9	40.5	47.1	30.3
N2	8	14.5	54.0	9.5	34.6
R3	17	27.6	72.8	14.0	7.5
N3	18	30.6	36.3	12.5	8.6
R4	119	58.1	71.5	10.7	2.5

Symbol definitions are the same as in Table 1.

although very scattered, show a clear trend away from a positive (normal) inclination to a negative (reversed) inclination. In both intervals the changes in the mean direction, which occur between 550°C and 650°C, are small compared to changes between the NRM and 550°C data.

Shown in Table 2 are the results of statistical analyses of the vectors removed during thermal demagnetization at 550°C. Note that for all the intervals the mean directions are fairly close to the present field direction ($D = 15°$, $I = 55°$). The differences between the mean directions of the removed vectors and the present field direction can be explained by the removal of small amounts of primary components during demagnetization in each of the intervals. Individual samples for which the removed vectors were closer to reversed than normal did occur in the R4 interval. Such samples in all cases occurred in short, stratigraphic intervals in which the demagnetized directions which were often anomalous, not being close to either the normal or reversed Moenkopi directions of *Helsley* [1969]. This may be due to the presence of secondary components of magnetization which are more stable than the primary components.

Figure 6 shows a stratigraphic plot of the demagnetized data beside a corresponding lithologic section. The directions shown are those after thermal demagnetization at either 550°C or 650°C, depending on whether the polarities were well defined at 550°C, as discussed above. The only significant effect of the additional demagnetization was a reduction in the scatter in the lower part of the lowest reversed interval (R1).

INTERPRETATION

In order to correlate the reversed and normal magnetozones observed at Bears Ears with those observed at Dolores by *Helsley and Steiner* [1974] we have assumed an approximate tem-

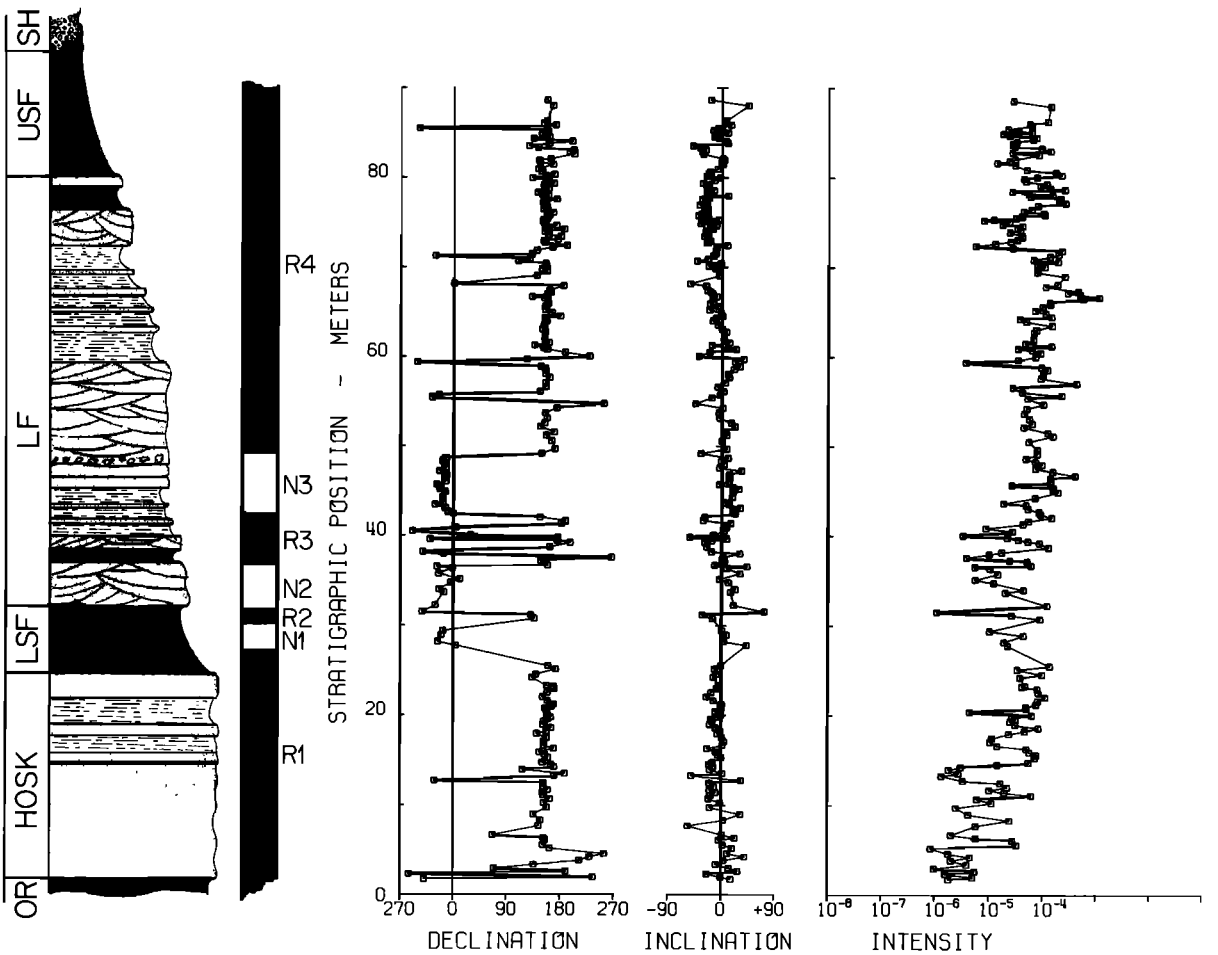


Fig. 6. Stratigraphic plot of thermally demagnetized data. Declinations are relative to geographic north. Intensities are total moments in emu for each of the samples. Also shown is a lithologic section and the proposed correlation with the magnetozones of *Helsley and Steiner* [1974]. Members of the Moenkopi Formation appearing in the left-hand column are defined as follows: OR, Organ Rock of the Cutler Formation; HOSK, Hoskinnini member of the Moenkopi Formation; LSF, Lower Slope Forming member of the Moenkopi Formation; LF, Ledge Forming member of the Moenkopi Formation; USF, Upper Slope Forming member of the Moenkopi Formation; and SH, Shinarump member of the Chinle Formation.

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poral equivalence of the Hoskinnini member at Bears Ears with the Tenderfoot member at Dolores. Such a correlation is supported by the work of *Stewart* [1959] discussed earlier. *Helsley and Steiner* [1974] showed that the lower two thirds of the Tenderfoot member at Dolores had a reversed magnetic polarity, which they termed R1. We have therefore correlated this reversed interval with the lowermost reversed interval at Bears Ears. Note that the lower portion of the R1 interval at Bears Ears (Figure 6) shows more scatter than the upper part. The data shown in Figure 6 for the lower part of R1 have been demagnetized at 650°C, as they showed even more scatter when demagnetized at only 550°C. The anomalous directions in the lower part of R1 all showed motion toward a reversed direction during demagnetization. Such behavior is probably due to the presence of secondary components of magnetization having directions close to the present field (see Table 2). Such secondary components may still be present in some of the samples even after demagnetization at 650°C.

Above the R1 interval is a fairly short normal interval which we correlate with the N1 interval at Dolores. All the samples in the N1 interval were demagnetized at 550°, 600°, and 650°C (the results shown in Figure 6 are for 650°C) to check the magnetic stability. None of these samples showed significant changes in their directions of magnetization with increasing demagnetization temperature. Above the N1 interval is a very short reversed interval consisting of only two samples which we have correlated with the R2 interval at Dolores. One of these samples (MH71OA) was subjected to the detailed stepwise thermal demagnetization discussed earlier (see Figures 4 and 5). The direction changes for this sample clearly indicate the presence of a stable reversed component at temperatures greater than 550°C. The behavior of the samples in the N2 interval above R2 is similar to that of those in the N1 interval. One of the pilot samples shown in Figures 3 and 4 (MH706B) is taken from this interval. The behavior of MH706B during demagnetization clearly indicates the presence of a stable normal component of magnetization. Unlike the other magnetozones, the N2 magnetozone coincides with a lithologic boundary, namely, a change from shale to crossbedded sandstones/siltstones. This coincidence may in part be due to the presence of an unconformity at the base of N2, which is also the top of the Lower Slope Forming unit (see earlier discussion). Such an unconformity would also explain the short length of the R2 interval (3 m compared to 20 m at Dolores).

The interval termed R3 in Figure 6 shows a considerable amount of scatter, especially in the middle portion. All samples in this interval were demagnetized at 550°, 600°, and 650°C to look for trends (the data in Figure 6 are for 650°C). In general, the R3 samples showed a trend toward a reversed direction with demagnetization (Table 1). However, two of the samples moved toward a normal direction. It is possible that these normal samples represent short events which were not recorded by the Dolores section. The 'anomalous' interval identified by *Helsley and Steiner* [1974] in the middle of the R3 interval at Dolores is not present at Bears Ears. This interval was either not recorded, owing to a local hiatus, or is obscured by secondary components of magnetization not removed by thermal demagnetization.

The N3 and R4 intervals are both well defined after 550°C demagnetization, apart from some isolated anomalous directions. As discussed in an earlier section, the anomalous directions occur in short intervals in which the directions of the components removed during demagnetization were closer to reversed than to normal directions. Additional demagnetization of these anomalous samples at 600°C and 650°C produced only small changes in their directions. We therefore conclude that these directions are probably due to the presence of stable secondary components which cannot be removed by thermal demagnetization.

DISCUSSION

Assuming that acquisition of the primary magnetization throughout the Moenkopi Formation was penecontemporaneous with deposition [*Baag and Helsley,* 1974*a*; *Elston and Purucker,* 1979], the temporal relationships between members of the Bears Ears section and the Dolores section studied by *Helsley and Steiner* [1974] can be established. The correlation is illustrated in Figure 7, which shows the two sections plotted on similar scales. In the figure the Hoskinnini member is assumed to be temporally equivalent to the lower part of the Tenderfoot member for the reasons discussed earlier. The Lower Slope Forming member is then shown to be equivalent to the upper part of the Tenderfoot and the lower part of the Ali Baba member. The length of the N1 magnetozone at Bears Ears suggests that the sedimentation rate was much lower than at Dolores, where the N1 magnetozone is more than 4 times as thick. Further up in the section the sedimentation rates appear to be more comparable. Although the top of the R4 magnetozone is not defined at Bears Ears, its minimum thickness is only slightly less than that of the R4 magnetozone at Dolores. It follows that the sedimentation rate at Dolores cannot be very much greater than that at Bears

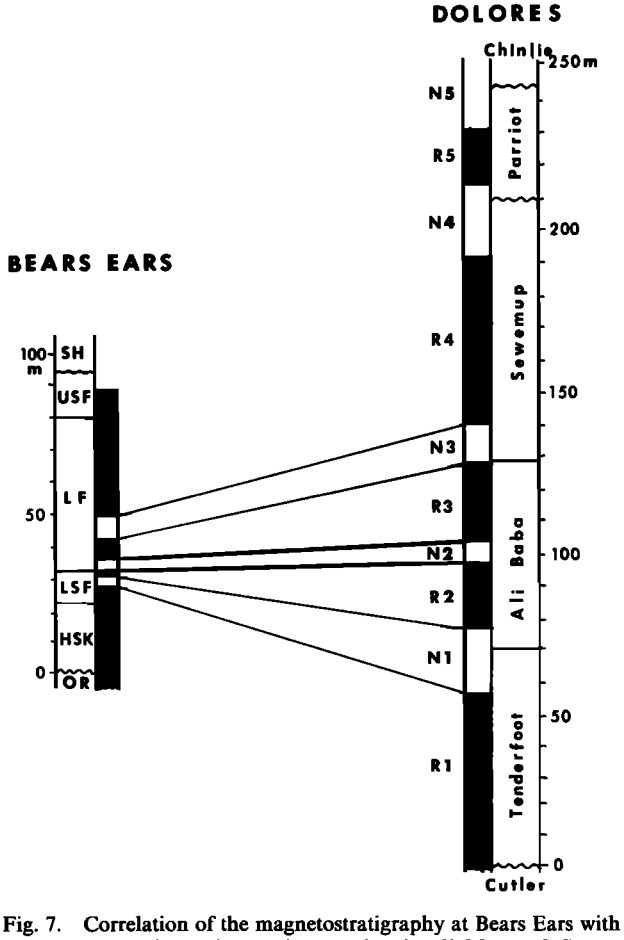


Fig. 7. Correlation of the magnetostratigraphy at Bears Ears with that observed in the Dolores River section by *Helsley and Steiner* [1974].

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Ears during the same interval. On the basis of the Bears Ears lithologies the temporal equivalents to the Upper Sewemup and Parriot members present at Dolores do not seem to be present at Bears Ears. These members were either not deposited or were removed by pre-Chinle erosion.

The paleomagnetic stability of the Bears Ears section is, in terms of the initial scatter of NRM directions, somewhat lower than the stability of the samples collected by *Helsley and Steiner* [1974]. The samples from the Dolores section gave well-clustered NRM directions which changed very little with thermal demagnetization [*Helsley,* 1969]. The reason for this variation in stability is probably lithologic as the coarse-grained massive sandstone units, which compose a large part of the Bears Ears section, are more common in southeastern Utah than in the salt anticline region to the north [*Stewart et al.,* 1972]. *Baag and Helsley* [1974*b*] have shown that the dispersion of NRM directions in subsurface samples of the Moenkopi generally increases with increasing grain size, which implies lower paleomagnetic stabilities in coarse-grained lithologies.

The main conclusions regarding temporal relationships between the two sections are therefore that the large difference in their thicknesses is because of (1) a lower sedimentation rate during the R1 and N1 intervals, (2) an erosional unconformity at the top of the Lower Slope Forming member, causing the loss of a large amount of the R2 magnetozone, and (3) lack of temporal equivalents to the Parriot and Upper Sewemup members that are present in Western Colorado. It is particularly striking to note that in the Bears Ears section the magnetozone boundaries occur within different members than corresponding boundaries seen in other sections of the Moenkopi. An example of this is the R1-N1 boundary, which at Bears Ears occurs in the Lower Slope Forming member. In the Dolores River section the lowest transition from reversed to normal occurs within the upper part of the Tenderfoot member (Figure 7). This lack of detailed parallelism in the physical stratigraphy supports the observations of *Blakey* [1973] that contemporaneous lateral variations in sediment type and environment are a characteristic feature of the Moenkopi Formation. At the same time the crossing of the lithologic boundaries by magnetozone boundaries is a strong argument against any correlation of magnetic polarity with local lithology.

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